

# ORIGINAL RESEARCH ARTICLES

## Impact of a Pharmacist-Led Warfarin Self-Management Program on Quality of Life and Anticoagulation Control: A Randomized Trial

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**Study Objective.** To evaluate the impact of a pharmacist-led warfarin patient self-management program on quality of life and anticoagulation control compared with management in a physician-led specialized anticoagulation clinic.

**Design.** Prospective, randomized, controlled, open-label trial.

**Setting.** Tertiary care academic medical center.

**Patients.** A total of 114 patients aged 18–75 years who were followed at a specialized anticoagulation clinic, had received warfarin for at least 6 months, and were expected to continue warfarin for a minimum of 4 months.

**Intervention.** All patients attended an educational session on anticoagulation provided by a pharmacist. Patients randomized to the self-management group (58 patients) also received practical training to use the CoaguChek XS device and a self-management dosing algorithm. Patients in the control group (56 patients) continued to undergo standard management at the anticoagulation clinic.

**Measurements and Main Results.** Patients completed a validated quality-of-life questionnaire and the validated Oral Anticoagulation Knowledge test at the beginning and end of the study. The quality of anticoagulation control was evaluated by using the time spent in therapeutic range. After 4 months of follow-up, a significant improvement in the self-management group was observed compared with the control group in four of the five quality-of-life topics ( $p < 0.05$ ). Improvements in knowledge were observed in both groups after the training session and persisted after 4 months ( $p < 0.05$  for all). The time spent in the therapeutic range (80.0% in the self-management group vs 75% in the control group,  $p = 0.79$ ) and in the extended therapeutic range ([target international normalized ratio  $\pm 0.3$ ] 93.2% in the self-management group vs 91.1% in the control group,  $p = 0.30$ ) were similar between groups.

**Conclusion.** A self-management warfarin program led by pharmacists resulted in significant improvement in the quality of life of patients receiving warfarin therapy as well as a reduction in the time required for anticoagulation monitoring, while maintaining a level of anticoagulation control similar to a high-quality specialized anticoagulation clinic.

**Key Words:** warfarin, anticoagulation, quality of life, pharmacist, self-management.

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Warfarin remains the most prescribed oral anticoagulant in North America.<sup>1</sup> Unfortunately, warfarin requires frequent monitoring of the international normalized ratio (INR) to maximize its efficacy and safety.<sup>2</sup> Although anticoagulants that do not require INR monitoring have been proposed as alternatives to warfarin, their efficacy and cost-effectiveness remain to be demonstrated in the context of optimal warfarin use and for many indications.<sup>3</sup>

With the emergence of portable monitoring devices, patients have the opportunity to perform INR self-testing and warfarin self-management. With the self-testing approach, patients measure their INR and report the results to their health care provider who adjusts the warfarin dose accordingly. With the self-management approach, patients measure their own INR and also perform their own dose adjustments. These novel approaches have yet to be widely adopted in North America.<sup>4, 5</sup>

Many studies have reported improvements in the quality of oral anticoagulation with self-testing and self-management approaches, with possible improvements in clinical outcomes.<sup>6, 7</sup> European data suggest that both approaches result in improvements in quality of life,<sup>8–13</sup> although data specific to self-management are not available in North America.

Finally, although extensive data have demonstrated that pharmacists in anticoagula-

tion clinics lead to improvements in the quality of anticoagulation therapy and monitoring,<sup>14, 15</sup> to our knowledge, no study has tested the impact of a pharmacist-led self-management program.

Thus, the objective of this study was to evaluate the impact of a pharmacist-led patient self-management program on quality of life and anticoagulation control in patients requiring chronic warfarin therapy compared with standard treatment in a physician-managed specialized anticoagulation clinic.

## Methods

### Study Design

In this prospective, randomized, controlled, open-label trial, patients were randomized in a 1:1 ratio to the self-management group, which used the CoaguChek XS (Roche Diagnostics Canada, Laval, Canada) device to monitor their INRs, or the control group, which consisted of standard management at the specialized anticoagulation clinic of the Montreal Heart Institute (Montreal, Canada). The validity of the CoaguChek XS compared with traditional laboratory testing has been reported previously.<sup>16, 17</sup> Institutional review board approval was obtained for the research protocol. The randomization list was generated by using permuted random blocks of sizes four and six. This list was generated using the PLAN procedure in SAS, version 9.1 (SAS Institute Inc., Cary, NC), by the Montreal Heart Institute Coordinating Center biostatistics department. All patients provided informed consent to participate in the study. The trial was registered at ClinicalTrials.gov (ClinicalTrials.gov registry number NCT01033279).

### Patient Population

Patients were eligible for the study if they were followed at the Montreal Heart Institute anticoagulation clinic, were receiving warfarin for at least 6 months, and were expected to continue warfarin for a minimum of 4 months. Patients were required to be aged 18–75 years, and their last two INRs had to be 1.5–4.0 for a target INR of 2.0–3.0, or 2.0–4.0 for a target

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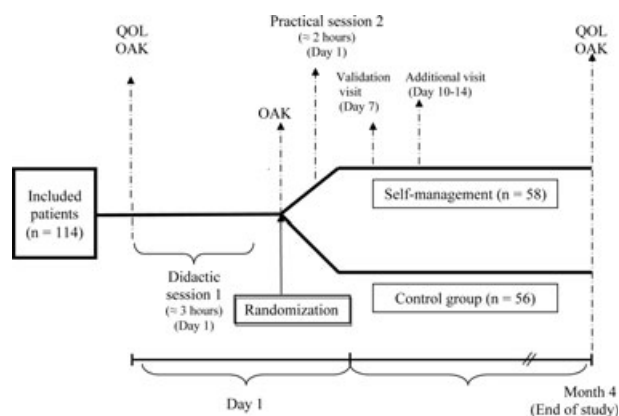
INR of 2.5–3.5. Patients were excluded if they had a hypercoagulable condition, including active cancer; had an active or recent major bleeding event in the last 3 months; experienced a recent cardiovascular event or surgery; or had a cognitive or physical impairment limiting their comprehension or motor skills.

### Study Protocol

On day 1, all patients attended a 3-hour educational lecture on anticoagulation provided by a pharmacist (Figure 1). During this didactic session, the patients received information on anticoagulation treatment and monitoring, as well as anticoagulation self-management. Patients completed a validated quality-of-life questionnaire<sup>9, 10</sup> at the beginning of the session, as well as the validated Oral Anticoagulation Knowledge (OAK) test<sup>18</sup> to test their knowledge about anticoagulation. Patients were then randomized to continue their management at the anticoagulation clinic (control group) or to switch to self-management (self-management group). Patients randomized to the control group received no further training. Patients randomized to the self-management group immediately received practical training to use the CoaguChek XS device and the self-management algorithm developed by the investigators (Appendix S1). This included instructions on

the frequency of INR monitoring, specific recommendations on what to do in the case of high or low INR, how and when to communicate with the pharmacists in the self-management program, how to use the device, and the patient's responsibility in the program. The session was concluded with clinical scenarios, during which patients had to apply their knowledge. They returned 1 week later to validate their use of the device and the algorithm. Patients who had difficulty using the device or algorithm at the second visit were invited to an additional visit. After this third visit, if the difficulties persisted, the patient would not have been allowed to undergo self-management.

On a weekly basis, patients in the self-management group monitored their INR and adjusted their dose of warfarin according to the algorithm. Through a voicemail message, patients were required to communicate their INR result and any adjustment performed. The patient was contacted if no telephone call was received on the expected day, or if an error in management occurred. If an INR fell outside the algorithm limits ( $\leq 1.5$  or  $\geq 4.5$  for a target INR of 2.0–3.0, or  $\leq 2.2$  or  $\geq 5.0$  for a target INR of 2.5–3.5), the dose was adjusted by the pharmacist. Each week patients also had to complete a pre-INR questionnaire to identify new factors that could affect their INR such as the addition of prescription or nonprescription drugs, natural health products, new health issues, missed doses, and changes in diet. In the presence of such factors, patients were required to inform the pharmacist of this information before making any dose adjustment. Patients in the self-management group recorded their INR, daily doses of warfarin, and adverse events. All patients were seen again after 4 months to complete the OAK test and the quality-of-life questionnaire. The INR results and dose adjustments from patients in the control group were collected directly from the anticoagulation clinic database.



**Figure 1.** Schematic of the study design illustrating the different aspects of the study after patient inclusion. The didactic session included information on general knowledge of oral anticoagulation. The practical session included training specific to the use of the CoaguChek XS device and use of the self-management algorithm. During the validation visit, the understanding of the technique of the device was validated, and a simulation exercise was performed to validate the understanding of the algorithm. OAK = Oral Anticoagulation Knowledge Test; QOL = Quality-of-Life questionnaire.

### Study End Points

The primary end point of the study was anticoagulation-related quality of life, measured with a previously developed and validated questionnaire, which includes 32 statements covering five sociopsychological topics: general treatment satisfaction, self-efficacy, daily hassles, psychological distress, and strained social network.<sup>9, 10</sup> Briefly, general treatment satisfaction measures

whether a patient feels more content with greater self-management of oral anticoagulation therapy. The self-efficacy portion of the questionnaire reflects patient self-confidence in dealing with challenging environmental demands by taking adaptive action. The daily hassles portion of the questionnaire is related to how the cost and benefit of the intervention affect the patient's daily life. General psychological distress provides information regarding the general psychological burden associated with a patient's oral anticoagulation. Finally, the strained social network focuses on the quality of the social support. Each domain is evaluated using a graded scale ranging from a score of 1 (total disagreement) to 6 (total agreement). Higher scores for self-efficacy and general treatment satisfaction, and lower scores for daily hassles, psychological distress, and strained social network are indicative of better quality of life.

Many secondary end points were evaluated. Knowledge regarding oral anticoagulation was compared between groups at baseline, after the didactic session, and at the end of the study using the OAK test.<sup>18</sup> The quality of anticoagulation control was evaluated using the time spent in therapeutic range, and the time spent within the extended therapeutic range (target INR  $\pm$  0.3) and outside the critical INR value (INR  $<$  1.5 or  $>$  5), all calculated with the Rosendaal method.<sup>19</sup> We recorded the average time required for patients to monitor their INR. For the self-management group, this consisted of the time spent measuring the INR with the device and performing dose adjustments. In the control group, it consisted of the average time required to travel to and from the anticoagulation clinic and the time spent at the clinic, estimated by patient self-report at the beginning of the study. We also recorded all adverse events. The severity of bleeding and thromboembolic events was classified as previously described.<sup>20</sup> Briefly, these complications were considered minor if they were not associated with costs or medical consequences (e.g., epistaxis), serious if they required treatment or medical evaluation (e.g., gastrointestinal bleeding), life-threatening (e.g., requiring surgical intervention), or fatal.<sup>20</sup>

### Statistical Analysis

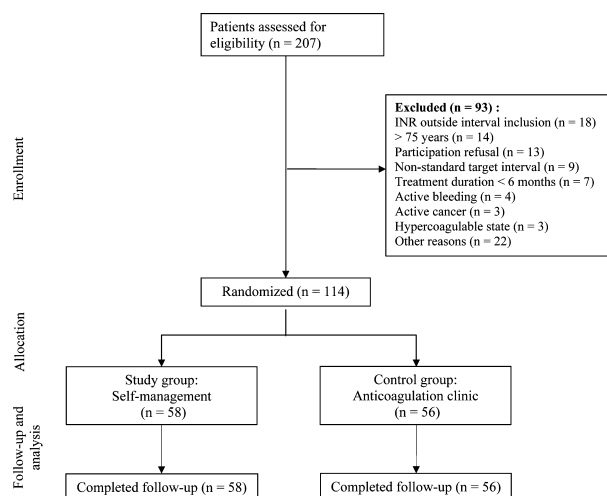
The sample size was based on previously published data.<sup>9</sup> To attain 80% power and an  $\alpha$  of 0.05 compared with a control group, we calculated that eight patients in the self-management

group were required to observe an improvement for the self-efficacy topic, 16 patients for daily hassles, 22 for strained social network, 94 for treatment satisfaction, and 184 for psychological distress. For feasibility reasons, we chose a study sample size powered to demonstrate differences in the first four aforementioned topics of quality of life. In order to compensate for a possible loss to follow-up, the number was increased to 120 patients. Enrollment was stopped after inclusion of 114 patients due to limited resources and because it was expected that more than the required 94 patients would complete the study.

Continuous values are presented as mean  $\pm$  SD when appropriate. Categorical variables are presented as numbers and percentages. A *t* test for independent samples was used to compare the mean values between groups. Changes from baseline within one group were calculated using a paired *t* test. For categorical variables, a  $\chi^2$  method was used. To compare the time spent in the therapeutic INR range or outside the critical INR range, the method previously proposed by Rosendaal was used.<sup>19</sup>

### Results

Between November 2009 and January 2010, 207 patients were screened for participation in the study. Of those, 93 were excluded; reasons for exclusion are presented in Figure 2. Hence, 114 patients were randomized: 58 to the self-management group and 56 to the control group. The last date of 4-month follow-up occurred on May 17, 2010. Baseline characteristics of the



**Figure 2.** Schematic of the patient selection process. INR = international normalized ratio.

study population are presented in Table 1. Although there was no statistically significant difference between the groups at baseline, there was a trend for a higher proportion of patients in the self-management group to have a target INR of 2.5–3.5 ( $p=0.06$ ). Patients consisted mainly of men (68%), with a mean  $\pm$  SD age of  $57.7 \pm 10.4$  years. The majority of patients received warfarin for atrial fibrillation or a prosthetic heart valve. Patients had been treated with warfarin for  $7.5 \pm 6.3$  years before enrollment. A total of 34.2% of the patients had a high-school diploma or less as the highest degree earned. No patients were lost to follow-up. In the self-management group, warfarin was discontinued after 3.3 months in one patient after a successful cardioversion.

### Quality of Life

Data on quality of life at baseline and follow-up were incomplete or missing for two patients

in the self-management group and three patients in the control group. Both groups were comparable with regard to quality of life topics at baseline (Table 2). After 4 months, treatment satisfaction significantly improved in the self-management group compared with the control group ( $p<0.001$ ). Daily hassles, psychological distress, and a strained social network all significantly decreased in the self-management group compared to the control group ( $p=0.024$ ,  $p=0.029$ , and  $p<0.001$ , respectively). Self-efficacy increased in both groups ( $p<0.05$ ), but the difference between the groups was not significant.

### Anticoagulation Control

The number of INRs measured/patient in the self-management group was greater than in the control group ( $17.4 \pm 1.3$  vs  $4.0 \pm 1.5$ ,  $p<0.001$  [Table 3]). Intervention by a pharmacist was required for 5.95% of the INRs performed in the self-management group because values fell outside the management algorithm. In the self-management group, after a dose adjustment using the algorithm was made after a nontherapeutic INR, the mean proportion of the next INRs/patient that were in the therapeutic range was  $73.9 \pm 24.5\%$ , whereas  $14.6 \pm 20\%$  were supratherapeutic and  $11.5 \pm 16.1\%$  were subtherapeutic. In the control group, these values were  $67.4 \pm 34.8\%$ ,  $14.0 \pm 24.9\%$ , and  $18.6 \pm 27.7\%$ , respectively. The mean proportion of therapeutic INRs/patient after a dose adjustment was not significantly different between groups ( $p=0.44$ ), nor was the total proportion of therapeutic INRs after a nontherapeutic INR in each group: 157 (64.1%) of 245 patients in the self-management group versus 43 (54.4%) of 79 patients in the control group ( $p=0.12$ ). As a post hoc analysis, we evaluated the number of errors that occurred when patients chose a dose for the upcoming week that led to an intervention by the pharmacist. We found four such errors, including two when the patient was in the therapeutic range and thus did not require a dose adjustment.

Similar anticoagulation control was observed in both groups with regard to the mean time spent within therapeutic INR range ( $p=0.79$ ), in the extended therapeutic INR range ( $p=0.30$ ), and outside the critical INR values ( $p=0.46$ ) (Table 3). Less time was dedicated to monitoring each INR in the self-management group compared with the control group ( $5.3 \pm 2.6$  vs  $158.0 \pm 67.8$  min,  $p<0.001$ ). Also, less total time was dedicated to anticoagulation monitoring

**Table 1. Baseline Characteristics of the Study Patients**

| Characteristic                           | Self-Management Group (n=58) | Control Group (n=56) |
|------------------------------------------|------------------------------|----------------------|
| Age (yrs)                                | $58.4 \pm 10.1$              | $57.0 \pm 10.9$      |
| Male                                     | 39 (67.2)                    | 39 (69.6)            |
| Highest academic degree completed        |                              |                      |
| College or university                    | 40 (69.0)                    | 35 (62.5)            |
| High school                              | 16 (27.6)                    | 21 (37.5)            |
| Grade school                             | 2 (3.4)                      | 0 (0)                |
| Treatment duration (yrs)                 | $8.0 \pm 6.6$                | $7.0 \pm 6.1$        |
| Time spent on anticoagulation (min/test) | $150.9 \pm 69.4$             | $157.0 \pm 68.2$     |
| Warfarin dose (mg/wk)                    | $43.2 \pm 17.0$              | $39.7 \pm 13.6$      |
| Target INR                               |                              |                      |
| 2.0–3.0                                  | 28 (48.3)                    | 37 (66.1)            |
| 2.5–3.5                                  | 30 (51.7)                    | 19 (33.9)            |
| Indication for anticoagulation           |                              |                      |
| Atrial fibrillation or flutter           | 25 (43.1)                    | 33 (58.9)            |
| Prosthetic heart valve <sup>a</sup>      | 30 (51.7)                    | 18 (32.1)            |
| Other                                    | 3 (5.2)                      | 5 (8.9)              |
| Treatment with antiplatelets agents      |                              |                      |
| Aspirin                                  | 10 (17.2)                    | 16 (28.6)            |
| Clopidogrel                              | 0 (0)                        | 1 (1.8)              |

Data are mean  $\pm$  SD or no. (%) of patients.

INR= international normalized ratio.

<sup>a</sup>Six patients in the self-management group and nine in the control group who had a prosthetic heart valve as their primary indication also presented with atrial fibrillation.

Table 2. Quality-of-Life Assessments in Self-Management and Control Groups<sup>a</sup>

| Quality-of-Life Domain         | Baseline Score        |               | End-of-Study Score    |               | Change from Baseline to End of Study |                          |                                        |
|--------------------------------|-----------------------|---------------|-----------------------|---------------|--------------------------------------|--------------------------|----------------------------------------|
|                                | Self-Management Group | Control Group | Self-Management Group | Control Group | Self-Management Group                | Control Group            | Self-Management Group vs Control Group |
| General treatment satisfaction | 4.1 ± 1.1             | 4.2 ± 1.2     | 5.3 ± 0.7             | 4.4 ± 1.2     | +1.3 ± 1.2<br>(p<0.001)              | +0.2 ± 1.1<br>(p=0.130)  | p<0.001                                |
| Self-efficacy                  | 5.0 ± 1.0             | 4.8 ± 1.1     | 5.4 ± 0.7             | 5.0 ± 1.0     | +0.4 ± 1.2<br>(p=0.014)              | +0.3 ± 1.1<br>(p=0.0047) | p=0.647                                |
| Daily hassles                  | 2.4 ± 0.7             | 2.4 ± 0.7     | 1.9 ± 0.6             | 2.2 ± 0.8     | -0.5 ± 0.6<br>(p<0.001)              | -0.2 ± 0.8<br>(p=0.162)  | p<0.024                                |
| Psychological distress         | 2.5 ± 1.0             | 2.5 ± 1.1     | 1.9 ± 0.7             | 2.2 ± 1.0     | -0.6 ± 1.0<br>(p<0.001)              | -0.2 ± 0.8<br>(p=0.033)  | p<0.029                                |
| Strained social network        | 2.0 ± 0.9             | 1.8 ± 0.8     | 1.4 ± 0.4             | 1.9 ± 0.8     | -0.6 ± 0.9<br>(p<0.001)              | +0.1 ± 0.7<br>(p=0.566)  | p<0.001                                |

Data are mean ± SD.

Scores range from 1 (total disagreement) to 6 (total agreement); higher scores for general treatment satisfaction and self-efficacy, and lower scores for daily hassles, psychological distress, and strained social network were indicative of better quality of life.

<sup>a</sup>Data were incomplete or missing for two patients in the self-management group and three patients in the control group.

Table 3. Anticoagulation Control

| Variable                                                                        | Self-Management Group | Control Group | p Value |
|---------------------------------------------------------------------------------|-----------------------|---------------|---------|
| No. of INR tests/patient                                                        | 17.4 ± 1.3            | 4.0 ± 1.5     | <0.001  |
| Percentage of time within target range <sup>a</sup>                             | 80.0 ± 13.5%          | 75.5 ± 24.7%  | 0.79    |
| Percentage of time within extended target range (target INR ± 0.3) <sup>a</sup> | 93.2 ± 7.9%           | 91.1 ± 13.3%  | 0.30    |
| Percentage of time outside critical INR value (INR < 1.5 or > 5) <sup>a</sup>   | 0.6 ± 1.8%            | 1.0 ± 3.8%    | 0.46    |
| Time spent monitoring each INR (min)                                            | 5.3 ± 2.6             | 158.0 ± 67.8  | <0.001  |
| Total time spent for anticoagulation monitoring during 4-mo follow-up (min)     | 99.6 ± 46.1           | 614.9 ± 308.8 | <0.0001 |

Data are mean ± SD.

INR = international normalized ratio.

<sup>a</sup>Measured by using the Rosendaal method.<sup>19</sup>

during the 4-month follow-up period (99.6 ± 46.1 vs 614.9 ± 308.8 min, p<0.0001).

(p=0.60) and after the 4-month follow-up period (p=0.98).

### Knowledge Related to Oral Anticoagulation Therapy

Twenty-three training sessions consisting of 1–9 patients each were held. Five patients required an additional training session because of difficulties using the device. No patient required a third session. Hence, all patients demonstrated proficiency in the use of the device and self-management algorithm in the self-management group. At the beginning of the study, both groups had a similar level of knowledge regarding anticoagulation use (Table 4). Compared with this baseline knowledge, significant improvement in knowledge was observed within each group after the training session (p<0.001 for both groups), which persisted after the 4-month follow-up period (p<0.05 for both groups). No significant difference was noted between groups in the knowledge acquired after the training session

### Adverse Events

The proportion of individuals presenting with an adverse event was similar between the groups (Table 5). The majority of events consisted of minor bleeding episodes, mainly epistaxis. Three major bleeding events occurred during the study, with two in the self-management group and one in the control group. The first one occurred in the self-management group in a patient who presented with an ocular hematoma after taking the wrong dose of warfarin. The INR increased to 5.4. The ocular hematoma resolved in 3 days without any intervention or hospitalization. The second major bleeding event in the self-management group was gastrointestinal bleeding caused by an acute exacerbation of chronic ulcerative colitis that required hospitalization. The major bleeding event that occurred in the control group was a case of hematuria after urologic surgery. No thromboembolic

Table 4. Oral Anticoagulation Knowledge Test Scores

| Baseline Score        | Change from Baseline to End of Didactic Session |                           |                                        |                           |
|-----------------------|-------------------------------------------------|---------------------------|----------------------------------------|---------------------------|
|                       | Self-Management Group vs Control Group          |                           | Self-Management Group vs Control Group |                           |
| Self-Management Group | Self-Management Group                           | Control Group             | Self-Management Group                  | Control Group             |
| 75.9 ± 14.0%          | +8.2 ± 10.4%<br>(p<0.001)                       | +9.4 ± 12.5%<br>(p=0.001) | +5.8 ± 12.5%<br>(p=0.002)              | +5.8 ± 13.3%<br>(p=0.002) |

Data are mean ± SD.

Table 5. Adverse Events

| Event                 | Self-Management Group (n=58) | Control Group (n=56) | p Value |
|-----------------------|------------------------------|----------------------|---------|
| All bleeding events   | 26 (44.8)                    | 23 (41.1)            | 0.75    |
| Bleeding              |                              |                      |         |
| Minor                 | 24 (41.4)                    | 22 (39.3)            | 0.88    |
| Major                 | 2 (3.4)                      | 1 (1.8)              | 0.59    |
| Life-threatening      | 0 (0)                        | 0 (0)                | —       |
| Thromboembolic events | 0 (0)                        | 0 (0)                | —       |
| Hospitalization       | 9 (15.5)                     | 6 (10.7)             | 0.47    |
| Mortality             | 0 (0)                        | 0 (0)                | —       |

Data are no. (%) of patients.

events or deaths were recorded. None of the 15 hospitalizations in the two groups were related to thromboembolic or hemorrhagic events or study interventions.

## Discussion

These results indicate that a self-management program led by pharmacists improves patients' quality of life and reduces the time required for anticoagulation monitoring, while maintaining control of anticoagulation similar to a specialized clinic.

Our results demonstrate a significant improvement in four of the five quality-of-life topics in the self-management group compared with management at the specialized anticoagulation clinic. These results support those of previous European studies,<sup>8–10</sup> with two of the studies reporting improvements in all five topics in patients self-managing their anticoagulation<sup>9, 10</sup> and the third study reporting improvement in four of the five topics.<sup>8</sup> More modest results were reported in another study.<sup>12</sup> To our knowledge, our study is the first quality-of-life study of anticoagulation self-management conducted in North America.

Of importance, the improvements in quality of life and time required for INR monitoring were observed with comparable control of anticoagulation. The level of anticoagulation control observed in both groups may help explain why, contrary to previous studies,<sup>6, 7</sup> we did not observe improvements in the time in the target INR range in the self-management group compared with the control group.<sup>6, 21</sup> The use of the Coaguchek XS device, along with the algorithm developed for the study, proved to be easy for the patients to comprehend and performed well. Only five patients (8.6%) required an additional visit to improve their skills using the

device, despite the fact that a third of our patients had a high school education or less. This is in agreement with The Home INR Study (THINRS) substudy, which evaluated patients' self-testing competency and concluded that a majority of them were able to achieve successful testing within 2–4 weeks.<sup>22</sup> A potential drawback of INR self-management is the higher number of tests required. However, the amount of time required by patients to monitor their warfarin therapy was lower in the self-management group than in the control group. This decrease in the time patients dedicate to their warfarin monitoring and the fewer anticoagulation clinic visits may have contributed to the improvements in quality of life observed. Recent data from a pilot study investigating the implementation of a new Internet-based tool for the management of anticoagulants indicated that these were two of the three most valued features of such systems by patients.<sup>23</sup>

An important longstanding issue regarding anticoagulation self-management is if patients actually benefit from self-management or if the improvements in anticoagulation control observed in previous studies were the result of the additional training that patients received.<sup>8, 9</sup> In these studies, the additional training was generally provided only to the self-management group. In our study, both groups attended a didactic session on anticoagulation, although the practical training for the use of the device was limited to the self-management group. Our results show that the improvement in knowledge evolved in a similar fashion in both the self-management and control groups. This indicates that self-management does not have a significant impact on knowledge retention.

Our study has limitations. First, the small number of patients precluded any evaluation of the impact of the self-management program on clinical events. Nevertheless, because the time spent within critical INR values was extremely low in the self-management group, one would expect a low risk of major events.<sup>24</sup> Furthermore, the level of anticoagulation control we observed was equal to or better than that in self-management studies that demonstrated reductions in such events.<sup>7</sup> The small sample size of our study may also have contributed to the fact that both groups were not completely identical at baseline. Another potential limitation of the study is related to the study population. The patients included in our study were receiving warfarin therapy for a mean of approximately

7 years. Therefore, our results cannot be extrapolated to individuals who have recently started warfarin. In addition, 80 (38.7%) of the 207 patients screened were not eligible, which limits the application of our findings to the population studied. Finally, the short duration of the study should be noted.

## Conclusion

Our study indicates that patients treated with warfarin on a long-term basis can be trained by pharmacists to manage their warfarin dosing requirements. This self-management leads to improvements in quality of life and reduces the time dedicated to INR monitoring while maintaining a quality of the anticoagulation comparable to that provided by a specialized anticoagulation clinic.

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## Supporting Information

The following supporting information is available in the online version of this article:

**Appendix S1.** Decision Algorithms for International Normalized Ratios of 2–3 and 2.5–3.5, and Weekly Coumadin Dose Adjustment Table

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